

Decision support summary derived from longitudinal vital sign trends; interpret in clinical context.

SECTION 1 — Clinical Intelligence: Signal Summary and Trajectory

Over 7 days of monitoring, 143 of 168 expected hours were recorded (85.1% coverage). 12 episodic events spanning 60 total hours were detected, including: Elevated Breathing Rate, High Heart Rate, Low Heart Rate, Very Low Heart Rate. The period showed 3 distinct phases. **Phase 1: Mixed Instability (Jun 23 to Jun 23):** Mixed episode pattern with average HR of 60 bpm. **Phase 2: Low HR + Elevated RR (Jun 24 to Jun 26):** Heart rate dropped to lows of 40 bpm with 2 episodes showing concurrent elevated breathing rate. **Phase 3: HR Surge (Jun 27 to Jun 29):** Heart rate surged with daily averages reaching 81 bpm and peaks near 109 bpm. The P5 to P95 heart rate spread was 51 bpm (47 to 98), indicating significant cardiac variability. 2 episodes showed concurrent low heart rate with elevated breathing rate, a coupling pattern that may reflect compensatory respiratory stress.

Trend: Progressively unstable vital sign pattern

Action: Provider review advised

Suggested Clinical Review Actions

- Phase 1: Mixed Instability (Jun 23 to Jun 23): Mixed episodes including Very Low Heart Rate. Correlate with clinical context, medication timing, and symptom assessment.
- Phase 2: Low HR + Elevated RR (Jun 24 to Jun 26): Very Low Heart Rate for 10 hours (min HR 40 bpm). Review heart rate lowering medications, check blood pressure, assess oxygen levels and fluid status due to concurrent elevated breathing rate.
- Phase 3: HR Surge (Jun 27 to Jun 29): High Heart Rate for 4 hours (avg HR 102 bpm). Evaluate for pain, infection, fever, dehydration, or heart rhythm change.
- Overall trajectory shows 3 unstable phases with 40 bpm variability. Consider increasing monitoring frequency and lowering escalation threshold.

SECTION 2 — High Priority Episodic Events

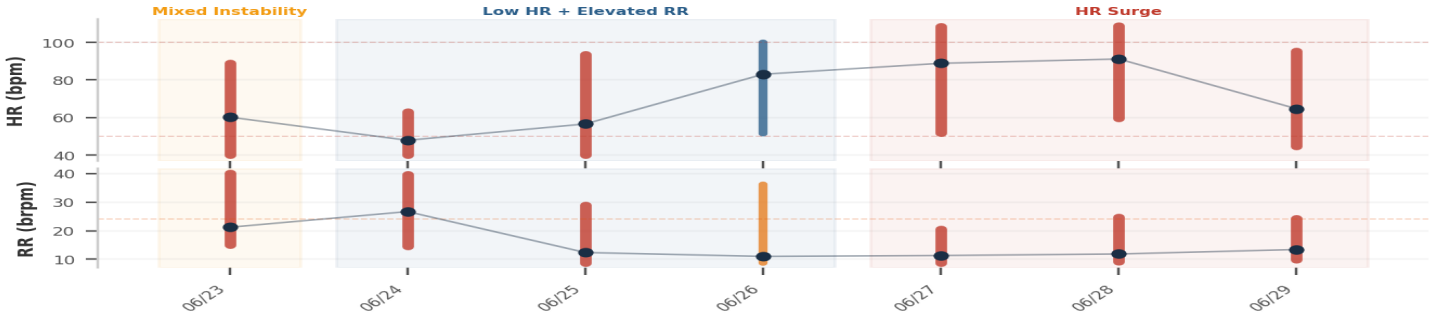
Event Type	Date / Time Window	Dur	Key Vitals	Clinical Concern
Very Low Heart Rate	06/24 00:00 to 06/24 00:00	10h	HR 47 Min 40 RR 40	Critical; urgent review advised (HR+RR coupled)
Low Heart Rate	06/24 00:00 to 06/24 00:00	11h	HR 47 Min 40 RR 40	Critical; urgent review advised (HR+RR coupled)
Elevated Breathing Rate	06/24 00:00 to 06/24 00:00	11h	HR 47 Min 40 RR 40	Sustained; consider provider review
Very Low Heart Rate	06/23 00:00 to 06/23 00:00	6h	HR 51 Min 40 RR 39	Sustained; consider provider review
Very Low Heart Rate	06/25 00:00 to 06/25 00:00	5h	HR 45 Min 40 RR 29	Sustained; consider provider review
Low Heart Rate	06/25 00:00 to 06/25 00:00	4h	HR 44 Min 40 RR 29	Sustained; review context

SECTION 3 — Statistics

Metric	Mean	Min	Max	P5	P95
Heart Rate Avg (bpm)	72.4	41.3	103.6	46.6	98.1
Heart Rate Min (bpm)	63.1	40.1	99.5	40.5	93.6
Breathing Rate Avg (brpm)	14.8	9.7	29.1	10.2	28.2
Breathing Rate Max (brpm)	21.8	11.2	40.2	12.5	38.7

SECTION 4 — Visual Trend Analysis

Daily Vital Sign Ranges (Phase shading: Mixed Instability; Low HR + Elevated RR; HR Surge)



Distribution (Red arrow = P5 to P95 range)

